Toward the Vectorization of Hyperspectral Imagery

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***Abstract*— Hyperspectral images (HSIs) can provide rich spectral–spatial information that has been widely utilized in many fields, such as national defense, mineralogy, and agriculture. Most of the recent HSI interpretation methods are conducted in the raster pattern, which results in high memory costs, amplification distortion, and difficulties in topological editing. To address this issue, a novel end-to-end vectorization framework is proposed, called as the HSI vectorization network (HSI-VecNet), which learns a vector representation from spectral–spatial information through cross-level interactions. Specifically, this framework integrates low-level geometry information and high-level semantic instance information, which consists of two branches: the HSI semantic instance segmentation (HSIS) and the spectral-spatial junction prediction (SSJP). The HSIS conducts the raster-based classification and extracts the semantic information of each object in the HSI. In addition, the SSJP exploits spectral–spatial information to predict the positions of junctions in the HSI. The instance information of each object and the relations of junctions are then fused to vectorize the HSI. To verify the effectiveness of the proposed method, four hyperspectral datasets are vectorially labeled. Experimental results on these datasets demonstrate that the proposed end-to-end HSI-VecNet outperforms existing post-process vectorization methods. Our model and datasets will be made publicly available at https://github.com/yyyyll0ss/HSI-VecNet.**

***Index Terms*— Convolutional neural network (CNN), cross-task spectral–spatial attention module (CSSAM), hyperspectral image (HSI) vectorization.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain both spatial information and spectral information, where each pixel has continuous spectral information. Thus, the interpretation of HSI is a crucial task in the fields of remote sensing and machine learning [1]. As a vital component in the application of HSI, the HSI classification is essential in various fields such as geology, ecology, environment, mining, agriculture, and forestry [2], [3]. It serves as a critical link, delivering valuable information and insights that inform decision-making in these fields.

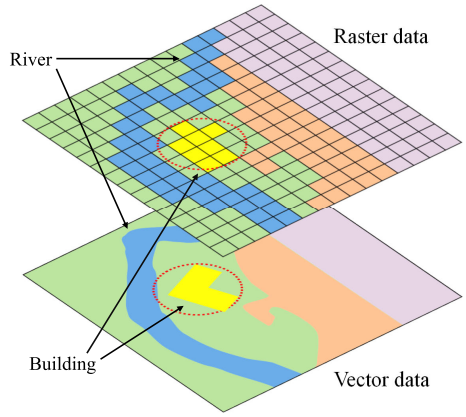


Fig. 1. Difference between raster data and vector data.

The task of HSI classification has been studied for decades [4], [5], [6], [7]. In the early stages, several methods [1] were employed for HSI classification, including spectral feature-based support vector machine (SVM) [4], random forest (RF) [5], multinomial logistic regression (MLR) [6], and so on. These methods only utilize spectral information to process the HSI and do not consider the spatial information among adjacent pixels. To enhance the accuracy of HSI classification, spatial information has been incorporated into the advanced pipelines [7], [8], [9]. Among deep learning-based methods, convolutional neural networks (CNNs), as the hierarchical spatial feature representation learning frameworks, have been widely used in HSI classification [10], [11], [12], [13], [14], [15].

Currently, HSI classification results are mainly presented as raster images, which are the direct outputs of pixel classification [1]. In general, the interpretation results in a raster form have some problems in the practical applications of geographic information. First, raster data requires storage for every pixel, including a lot of useless background pixels, leading to excessive storage demands, which is difficult to adapt to the limited storage space of mobile devices. Second, the raster classification results are not suitable for the subsequent graphics editing operations [16], such as moving, copying, attribute modification, and topology solving. Moreover, enlarging the raster images will result in a sawtooth effect on the outline of the objects. Fig. 1 illustrates the difference between raster and vector data. In brief, most remote sensing applications that involve target extraction require extraction results in vector format rather than rasterized masks.

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Existing researches [17], [18] on vectorization involve a multistep process: 1) generating the instance-level rasterized segmentation masks from the input his; 2) vectorizing the mask through heuristic post-processing; and 3) simplifying the result to minimize vertex redundancy. Among them, generating segmentation masks, as a key subproblem, has been significantly promoted by deep neural networks (DNNs) in recent years [19]. It is easy for us to obtain high-precision segmentation masks through HSI classification algorithms. After that, the transformation of vectors can be realized using Douglas–Peucker (DP) and other contour simplification algorithms [20]. However, these simplified algorithms rely too much on manual adjustment of parameters, and vector junctions tend to be scarce or redundant. Therefore, Chen et al. [21] proposed a two-stage model for optimizing the position of vector junctions, resulting in vectors with more precise locations. Recently, Zorzi et al. [22] introduced a method that first predicts vector junctions and then employs graph neural networks (GNNs) to learn how to connect junctions and obtain concise vector results. Toward the vectorization of HSI, the existing vector extraction methods cannot be directly applied to the HSI datasets. Unlike optical remote sensing images, HSIs have less labeled data and are more informative in spectral dimensions. The previous approaches cannot exploit the rich spectral information of HSI and require plenty of images for training. In general, the researches on vectorization are primarily focused on visible images, with limited studies on vectorizing HSI.

In this article, we propose the HSI vectorization network (HSI-VecNet) that learns a vector representation from spectral–spatial information through cross-level interactions. Existing post-process-based vectorization methods have the problems of the inaccurate position of vector junctions and the unreasonable number of junctions. To obtain accurate junction positions, we utilize the continuous spectral sequence of subtle changes in the HSI to regress junction positions. The proposed method has a two-branch structure, which can simultaneously predict the results of HSI classification and the location of junctions. Meanwhile, we employ the cross-task spectral-spatial attention module (CSSAM) to facilitate information interaction between the two branches. Practically, the CSSAM combines high-level semantic information from the classification task with low-level geometric information from junction prediction. In the inference stage, the model extracts the contours of each instance from the classification result and uses it to connect junctions for constructing vectors. Compared with the methods based on post-processing, the HSI-VecNet proposed in this article can make the position of vector junctions more accurate, and optimize the number of vertices globally without excessive manual adjustments. To validate the effectiveness of our proposed method, we relabeled the Indian Pines, Salinas Valley, Wuhan University HSI LongKou (WHU-Hi-LongKou), and Wuhan University HSI HongHu (WHU-Hi-HongHu) datasets, and the areas of different land types are labeled in vector form. Experiments show that the method proposed in this article can significantly improve the quality of HSI vectorization and obtain more concise and accurate results.

Our contributions are summarized below.

1) We introduce the vectorization task into HSI processing for the first time and achieve promising results on the Indian Pines, Salinas Valley, WHU-Hi-LongKou, and WHU-Hi-HongHu datasets.
2) We present a novel approach for HSI vectorization, which learns a vector representation from spectral–spatial information through cross-level interactions, concurrently performing junction prediction and instance extraction to achieve end-to-end vectorization.
3) We devise the CSSAM to combine semantic information and geometric knowledge, which leverages the interaction of spatial and spectral information between the two branches to reduce the reliance on the training set and enhance the vectorization performance.

The rest of this article is structured as follows. Section II summarizes the previous works on HSI classification and vectorization in remote sensing images. In Section III, the proposed HSI-VecNet is introduced in detail. Section IV introduces the improvement of the dataset and all the experimental results and discussions. Finally, Section V summarizes the article and presents suggestions for future researches.

II. Related Works

HSIs have been extensively studied in classification [1]. Previous works have used spectral [23] or spectral–spatial information [24] to resolve the category of each pixel, which achieved high accuracy at the pixel level. But, the classification results in raster form cannot retain the instance information precisely and cause distortion when the image is enlarged. This article mainly focuses on the HSI vectorization method based on deep learning. Recent HSI classification methods based on deep learning and vector extraction methods of visible images in remote sensing are reviewed in this section.

A. Deep Networks for HSI Classification

Recent deep learning based HSI classification can be roughly divided into spectral-based methods [23], [25] and spectral–spatial based methods [26], [27]. With the development of the HSI analysis techniques, the analysis targets gradually evolved from individual pixels to patches and even larger scales [24], [28], [29]. So, we discussed the different approaches along this line.

Spectral-based methods mainly utilize continuous spectral information to extract features from individual pixels [23], [25]. Chen et al. [7] proposed a stacked autoencoder to extract spectral features from pixels. Mou et al. [23] employed a recurrent neural network (RNN) to analyze the relationship between different spectral bands. Hang et al. [28] proposed a cascaded RNN to eliminate redundant spectral bands. Wu and Prasad [25] exploited the 1-D convolutional network and RNN to extract contextual features. In general, spectral-based methods constantly use stronger feature extractors to learn the spectral information of individual pixels to improve classification accuracy. The spectral-based techniques only focus on the relationship between spectral bands, ignoring the relation between different pixels on the spatial level [30].

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Spectral-spatial methods simultaneously utilize the spectral information of different bands and the spatial information between different pixels [26], [27]. How to make full use of spatial contextual information is always the research goal of these methods. Therefore, we will discuss its evolution from a single pixel to a larger spatial scale. Chen et al. [12] proposed a 3-D convolutional neural network (CNN) combined with L_2 regularization, which learned from spectral–spatial information but only used the adjacent pixels in a fixed range. To enlarge the spatial receptive field, Lee and Kwon [31] utilized kernels of different sizes to extract contextual information from adjacent pixels. Since resizing the kernels artificially cannot be applied to more complex scenarios, Zhu et al. [32] employed a deformable convolution to extract features to enhance the performance. Besides, Roy et al. [24] utilized 2-D/3-D convolutions to build a hybrid network. To obtain spatial information beyond adjacent pixels, Hong et al. [27] employed graph convolutional network (GCN) to model long-range spatial relations in the HSI. Zhang et al. [33] leveraged six CNN branches to capture contextual interactional features from diverse regions. Recently, more researchers have focused on how to efficiently utilize a broader range of information. Hong et al. [29] proposed SpectralFormer to model sequence attributes of spectral bands, which can convey spectral information from shallow to deep layers and better integrate spatial information. Roy et al. [34] designed a new morphological transformer that uses spatial and spectral morphological convolutions to focus on target global announcements via dilation and erosion operators. In addition, Yang et al. [35] proposed a method to directly generate multiscale regions to break the interaction gap between local areas and further environments. In general, the spectral–spatial HSI classification methods have achieved a high precision [29] and gradually developed from local information modeling to overall information modeling [32], [35]. However, the results of HSI classification often do not have the attributes of the instance level, which is not conducive to the subsequent application of classification results [36]. Hence, using vector extraction in HSI to analyze the instance-level attributes of targets is one of the ways to realize more intelligent HSI interpretation.

B. Vector Extraction of Remote Sensing Images

The typical HSI classification and segmentation networks are easy to train and generate accurate classification masks [35], but most remote sensing applications that involve building layers require classification data in vector format rather than rasterized masks [37]. Compared with common raster data, vector data have natural advantages in object representation [21], which can more accurately represent the shape features of the target and require less memory consumption. Currently, in the literature, vector extraction methods of remote sensing images can be divided into two categories. The first category includes methods that mainly achieve vectorization by regularization of segmentation mask [38]. In contrast, the methods of the second category [22] directly learn the components of vectors, including junctions, lines, order of connection, etc.

The first category includes methods that mainly focus on the regularization of segmentation masks. In building extraction task [38], the authors first used the DP algorithm [20] to simplify the detected boundaries, thus generating vectors using the binary mask generated by Mask R-CNN [39]. This was the first attempt at vectorizing the segmentation masks by post-processing, but the vector accuracy was not high enough. After that, Chen et al. [40] exploited a relative angle gradient transform to regularize the mask by quantizing the histogram of building boundaries. Zorzi et al. [41] connected three different networks in series for target extraction and vectorization to improve the vector accuracy further. They used three CNNs to generate segmentation masks, regularize segmentation results, and detect junction points. These points will be connected according to the regularized boundary to obtain the final vector result. Recently, the framework field learning (FFL) method proposed by Girard et al. [42] showed better performance, which encodes useful boundary information through vector fields and guides the formation of vectors. In general, errors will accumulate in the pipeline of post-processing-based forms, which may deteriorate the final vector results.

The second category is represented by methods [22], [43] that directly learn the representation elements of vectors and use these to construct the vectors, usually including semantic information, junction locations, junction connection order, etc. For example, some networks used RNN to extract vectors point by point, such as polygon-RNN [44] and polygon-RNN++ [45]. PolyTransform [46] initialized a vector polygon for each object instance and uses transformer network [47] to refine vertex positions. In remote sensing images, PolyMapper [48] was proposed to predict the vertexes of buildings and roads by an RNN. Since using the RNN to learn the connection order is time consuming, Chen et al. [21] proposed a two-stage network that predicted the segmentation mask to obtain a sparse contour representation and used a network to optimize vector junctions. To explore how to build vectors more efficiently, PolyWorld [22] tried to predict vector junctions and employed a GNN to learn how to connect junctions. This method can improve the number of vector junctions and the accuracy of position but will cause excessive calculation costs on large-scale images due to the limitation of the GNN. Recently, Xu et al. [43] presented a hierarchical supervision learning scheme, which concurrently uses points, lines, and regions information to construct the vectors and obtains concise vector results. The second kind of method can achieve a higher quality of vectors than the post-processing-based methods. Designing more efficient and straightforward modeling methods is the leading research direction in the future.

Toward the vectorization of HSI, the existing vector extraction methods cannot be directly applied to the HSI datasets. Unlike optical remote sensing images, HSIs have less labeled data and are informative in spectral dimensions. The approaches above cannot leverage the rich spectral

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information of HSI and require plenty of images for training. Given the limitations of current methods in HSI vectorization, it is meaningful to propose an end-to-end HSI vectorization method for a better interpretation of HSI.

III. Methods

Although vectorization has attracted some attention in the field of remote sensing image processing, the researches on vectorization of HSI are still few. Especially, the existing methods are difficult to be directly applied to HSI. Generally speaking, training and test sampling usually take pixel as the basic unit for HSI datasets, which cannot be applied to the natural image-based vectorization methods. Therefore, we propose a vectorization method that is used for HSI.

A. Framework Overview

The structure of the HSI-VecNet is shown in Fig. 2. The whole vectorization process can be divided into feature extraction, cross-task spectral–spatial interaction, and the final vector assembly.

In the feature extraction stage, we use the full convolutional network (FCN) to extract the valuable information of HSI and get the multiscale feature maps shared by subsequent tasks. Due to the low spatial resolution of HSI, the downsampling operation will lead to the loss of low-level detail information. So, we use the high-resolution network (HRNet) [49] as our backbone network to attain the high-resolution feature maps.

After obtaining the feature maps, we employ two branches to achieve the semantic instance segmentation and junction regression, respectively. Semantic instance segmentation requires the model to learn high-level semantic information and classify different targets accurately. Unlike the typical HSI classification method, semantic instance segmentation also needs to consider the background class. Furthermore, an excellent vectorization result also contains high-precision junction locations, which requires the model to have the cognitive ability on low-level geometric information. To achieve these two sub-tasks in one stage, we design two branches to complete the two tasks, respectively. One is for classifying the semantic categories of different pixels in the image and the other is for returning the positions of various junctions. To improve the performance of the model, we also exploit the CSSAM to realize the spectral–spatial information interaction between the two branches.

For the vector assembly, we utilize category classification and junctions' regression results to construct the vector. Inspired by the scheme used by Xu et al. [43] in building vector extraction, we design a multiclass vectorization scheme that extracts the target contour from the segmentation result, then uses the connection sequence contained in the outline to match the nearest junctions one by one, and finally obtains the vectorized segmentation results.

B. Spectral-Spatial Cross-Task Interaction

In the two branches, the classification methods have been relatively mature [14], [24], [29], and there is little research on the junction location regression of HSI. So, we propose SSJP to achieve junctions position regression. In this part, we will introduce how to implement junctions' regression and how to realize cross-task interaction by CSSAM.

CSSAM realizes the combination of semantic information and geometric details by exchanging information between the two branches. F_{cla} and F_{ver} represent the features of the classification and junctions regression tasks, respectively. Since there is a specific correlation between the features of the two branches, we use CSSAM to emphasize the shape correctness of classification results and semantic consistency in junction learning. Inspired by the success of bottle attention module (BAM) [50], we use parallel structures to learn the critical features from both spatial and spectral perspectives, as shown in Fig. 3. In contrast, due to the intense diversity of HSI in spectral dimensions, different devices have different band numbers. In spectral attention, we refer to the 1-D convolution designed with the adaptive size of convolution kernel in efficient channel attention (ECA) [51].

The detailed structure of CASSM is illustrated in Fig. 3. For the two given feature maps $F_{\text{cla}} \in \mathbb{R}^{C\times H\times W}$ and $F_{\text{ver}} \in \mathbb{R}^{C\times H\times W}$, we use $F \in \mathbb{R}^{C\times H\times W}$ to represent the sum of two features. CASSM infers a 3-D attention map $M(F) \in \mathbb{R}^{C\times H\times W}$. The refined feature map F' is computed as

$$F' = (F_{\text{cla}} + F_{\text{ver}}) + (F_{\text{cla}} + F_{\text{ver}}) \otimes M(F_{\text{cla}} + F_{\text{ver}}) \tag{1}$$

where $\otimes$ denotes elements-wise multiplication. $M(F)$ consists of two parallel branches. We use global average pooling in the spectral attention module to get specific feature responses in each channel. Subsequently, a 1-D convolution module is used to extract the channels of interest. The size of the convolution kernel is set adaptively depending on the dimension of the given input channel that is proposed in ECA [51]. Finally, we get the spectral attention vector $F_c \in \mathbb{R}^{C\times H\times W}$ through an activation function. In summary, spectral attention is calculated as follows:

$$M_c(F) = \sigma(\text{C1D}(\text{GAP}(F))) \tag{2}$$

where σ represents sigmoid function, C1D represents 1-D convolution, and GAP represents global averaging pooling.

Spatial attention is used to emphasize or suppress features of different spatial locations. First, we use 1×1 convolution to reduce the input feature dimension to $\mathbb{R}^{C\times H\times W}$. After this, we apply two 3×3 convolutions to maintain a large receptive field, which makes better use of the context information. Finally, 1×1 convolution and batch normalization are used to return the dimension to the previous level. The calculation process is expressed as follows:

$$M_s(F) = \text{BN}\left(f_3^{1\times 1}\left(f_2^{3\times 3}\left(f_1^{3\times 3}\left(f_0^{1\times 1}(F)\right)\right)\right)\right) \tag{3}$$

where f denotes a convolution operation, BN denotes a batch normalization operation, and the superscripts represent the convolutional filter sizes. There are two 1×1 convolutions for channel reduction and recovery. The intermediate 3×3 dilated convolutions are applied to aggregate contextual information with a larger receptive field.

In the junction prediction task, we use junction classification results of pixels and offset to build the vector location instead

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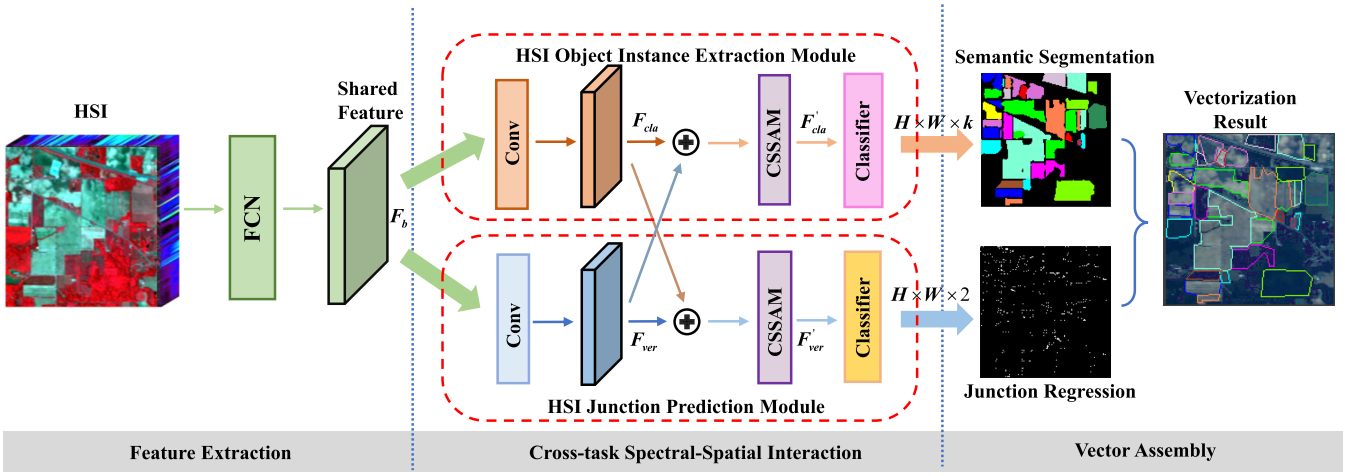


Fig. 2. Overview of HSI-VecNet. This method consists of the semantic instance segmentation branch and the junction prediction branch, which directly learns the components of the vector and aggregates the two results to get the vector result. At the same time, CSSAM realizes the information exchange between semantic features and geometric features, and improves the quality of vectors.

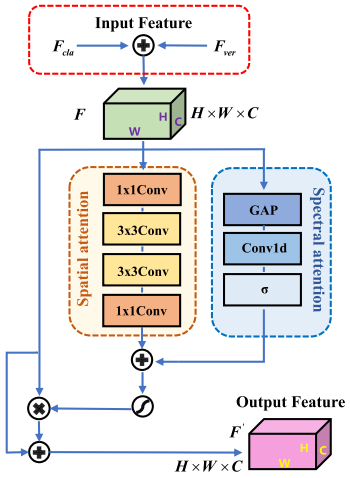


Fig. 3. Cross task spectral–spatial attention module.

of directly predicting. A convolution layer is used to learn the classification result C and the offset map O. The loss functions are cross entropy loss and l_1 loss by

$$\begin{aligned} L_{\text{ver}} &= L_{\text{vcla}} + L_{\text{voff}} \\ &= \text{BCE}\big(C, C^*\big) + l_1\big(O \cdot C^*, O^* \cdot C^*\big) \end{aligned} \tag{4}$$

where C^* is the ground truth of junctions and O^* is the short-range offsets that are in the range of $[-0.5,0.5]$ for the target pixels.

In the classification task, S represents classification results and S^* represents the label. BCE loss is used to calculate classification results of different categories, which is computed as

$$L_{\text{cla}} = \text{BCE}\big(S, S^*\big). \tag{5}$$

In summary, the total loss function consists of category classification loss, junction classification loss, and junction offsets loss. To better train the neural network, we use preset loss function factors to balance different losses in the total loss function by

$$L_{\text{total}} = \lambda_{\text{cla}} \cdot L_{\text{cla}} + \lambda_{\text{vcla}} \cdot L_{\text{vcla}} + \lambda_{\text{voff}} \cdot L_{\text{voff}} \tag{6}$$

where λ_{cla}, λ_{vcla}, and λ_{voff} are the loss factors of category classification, junctions classification, and junctions offsets. In our experiment, we set $\lambda_c = 4$, $\lambda_{\text{vcla}} = 8$, and $\lambda_{\text{voff}} = 0.25$ according to their loss magnitudes. We will introduce this in detail in the ablation experiment section.

C. Vector Assembly

Benefiting from the cross-task information interaction, the HSI segmentation results have sharper contours. However, due to the limitation of raster form, the vector obtained by regularizing the mask still cannot meet the needs of cartographic applications. To solve this problem, we can use the classification and junctions' prediction results to construct the vectors. Inspired by the mask-and-vertices attraction (MaV-Attr) scheme used by Xu et al. [43] in the vector extraction of buildings, we employ a multiclass vectorization scheme in the task of HSI vectorization as in Fig. 4.

We process the targets of different categories individually for the given HSI junction classification results C and junction offsets O. For the classification results of the same category, we first divide the classification mask S into n instances by calculating pixel connectivity, represented as $V_1, \ldots, V_n$. For every instance, we track all contour pixels to obtain the initial vectors that are redundant in junctions, expressed as $\partial V_i^{\text{init}} = [x_1^i, \ldots, x_r^i, x_{r+1}^i = x_1^i]$, where r is the number of boundary pixels of V_i. After obtaining the initial vectors, local non-maximum suppression (NMS) is used to simplify the prediction junctions C. A max-pooling layer with a kernel size of 3×3 implements the local NMS. Furthermore, the junctions with a higher classification score than 0.5 are retained and refined by the offset vectors for the sub-pixel level, denoted by $P = [p_1, \ldots, p_m]$.

In the next step, we operate the simplified prediction vector points P to match V_i^{init} for all categories. For the ith initial

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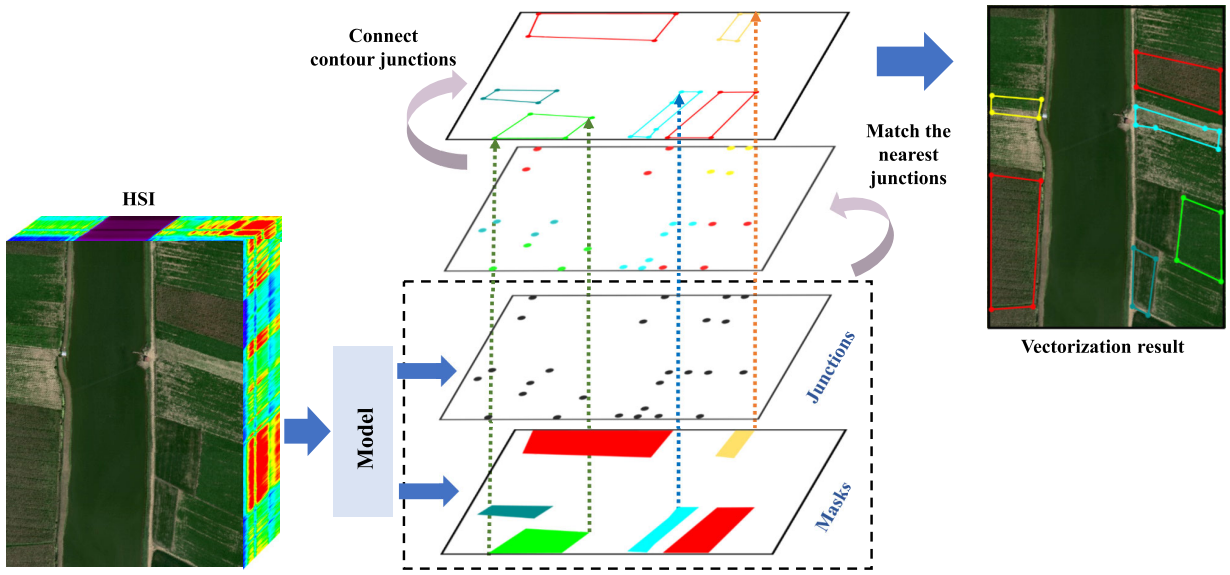


Fig. 4. Multi-class HSI vectorization scheme. Vectors are generated through matching masks with their nearest junctions.

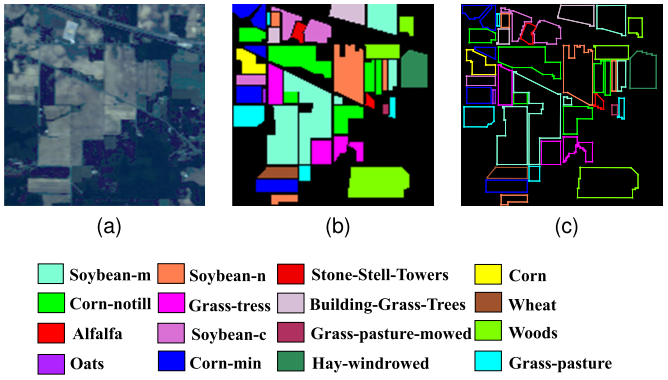


Fig. 5. Indian Pines dataset. (a) Three-band false color composite. (b) Original ground truth. (c) Vector ground truth.

vector V_i^{init}, we match all points to their nearest prediction junctions $p_{d(k,i)} \in P$ by computing the Euclidean distance by

$$d(k,i) = \arg_j \min_{v_j \in P} \left\| x_k^i - p_j \right\| \tag{7}$$

$d(k,i)$ represents the index of the point in P, which is closest to the kth contour point in the ith initial vector. Then, we go through all contour points x_k^i and get the corresponding $d(k,i)$ of all contour points. If two $d(k,i)$ are equal, we reserve the one with the minimum distance. After processing all contour pixels, the number of the retained points in V_i^{init} is the same as that of junctions successfully matched in P. With the connection order of V_i^{init}, the junction p_j in P is connected one by one to obtain the final vector result $P_{\text{final}} = \{p_1, \ldots, p_n\}$.

IV. Experiments

A. Dataset Description

HSI classification based on deep learning [11] is in a popular stage, but the common HSI datasets are all in raster form. At present, there is no research on HSI vectorization and available datasets, which restricts the development of HSI vectorization technology. Therefore, constructing the benchmark datasets for HSI vectorization is conducive to the intelligent interpretation of HSI. To this end, we selected four popular HSI datasets for relabeling and constructed the benchmark datasets for HSI vectorization.

1) Indian Pines: Indian Pines dataset was captured by the airborne visible infrared imaging spectrometer (AVIRIS) sensor in Northwestern Indiana, USA. This dataset consists of 145 × 145 pixels and 224 spectral–spectral bands in the wavelength range 0.4–2.5 μm. The ground truth of each pixel is designated into 16 classes and is not all mutually exclusive. By removing bands covering the bands of water absorption, [104–108], [150–163], and 220, the overall number of bands is reduced to 200. The result of the vector notation is shown in Fig. 5.

2) Salinas Valley: The Salinas Valley image was collected by the AVIRIS sensor over Salinas, California, which consists of 512 × 217 pixels with a spatial resolution of 3.7 m per pixel and 224 spectral bands. The ground truth contains 16 classes. The result of the vector notation is shown in Fig. 6.

3) WHU-Hi-LongKou: The WHU-Hi-LongKou dataset was acquired in Longkou Town, Hubei Province, China, with an 8-mm focal length Headwall Nano-Hyperspec imaging sensor equipped on a DJI Matrice 600 Pro (DJI M600 Pro) unmanned aerial vehicle (UAV) platform. The UAV flew at an altitude of 500 m, the size of the imagery is 550 × 400 pixels, there are 270 bands from 400 to 1000 nm, and the spatial resolution of the UAV-borne hyperspectral imagery is about 0.463 m. Details of each category and dataset division can be seen Fig. 7.

4) WHU-Hi-HongHu: The WHU-Hi-HongHu dataset was acquired in HongHu City, China, in 2017, by an UAV platform [52]. The experimental area is a complex agricultural scene with many types of crops. We intercepted the region with a size of 410 × 478 pixels. There are 270 bands with wavelengths ranging from 0.4 to 1 nm. The available

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TABLE I

COMPARISON RESULTS OF DIFFERENT HSI VECTORIZATION METHODS ON THE INDIAN PINES DATASET, INCLUDING PER-CLASS IOU, C-IOU, AND POLIS

Category	SpectralFormer [29] + DP			MorphFormer [34] + DP			HRNet [49] + DP			HSI-VecNet		
	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓
class1	0±0	0±0	—	0±0	0±0	—	0±0	0 ±0	—	0±0	0±0	—
class2	81.05±5.85	62.31±7.21	1.451±0.152	80.23±6.32	60.32±5.85	2.162±0.152	79.17±3.42	61.79±5.51	1.252±0.102	82.99±3.36	78.4±5.27	0.532±0.023
class3	80.12±7.60	66.39±7.56	1.232±0.086	81.93±6.21	66.96±8.20	1.363±0.085	81.11±7.69	65.57±4.50	1.412±0.152	81.00±6.50	75.52±3.96	0.547±0.062
class4	74.56±11.34	48.96±13.36	1.675±0.233	71.45±9.36	42.31±10.23	1.495±0.163	74.06±10.08	44.22±11.58	2.036±0.201	87.47±5.32	78.25±9.59	0.706±0.043
class5	70.36±8.21	45.74±8.22	1.625±0.102	71.32±7.68	42.20±8.25	1.235±0.142	72.20±7.33	46.13±6.33	1.675±0.189	83.09±6.19	77.14±6.33	0.702±0.053
class6	74.22±1.80	30.25±4.16	0.679±0.082	74.23±4.06	29.63±4.95	1.695±0.056	75.61±8.14	32.11±3.67	1.265±0.152	79.86±4.64	65.2±3.50	1.021±0.069
class7	0±0	0±0	—	0±0	0±0	—	0±0	0±0	—	0±0	0±0	—
class8	93.72±4.65	81.36±7.63	0.832±0.078	92.30±3.28	61.79±5.96	1.014±0.062	94.02±5.38	80.34±6.36	0.784±0.063	94.02±3.52	80.34±5.52	0.952±0.052
class9	0±0	0±0	—	0±0	0±0	—	0±0	0±0	—	0±0	0±0	—
class10	79.17±2.65	66.03±10.63	0.425±0.034	80.36±3.62	62.50±8.95	0.531±0.052	81.61±2.68	68.05±11.32	0.665±0.086	81.61±1.37	68.05±2.29	1.030±0.085
class11	83.92±3.96	75.61±12.21	0.632±0.056	86.63±2.56	72.65±9.36	0.449±0.042	88.58±6.98	76.85±13.56	0.754±0.063	88.58±1.83	76.85±9.59	0.512±0.031
class12	82.16±4.87	70.36±5.23	0.625±0.652	82.74±4.20	80.26±5.28	0.369±0.047	83.73±3.33	73.06±2.83	0.524±0.025	83.73±3.62	73.6±4.36	0.562±0.062
class13	86.53±12.20	80.22±12.63	0.495±0.032	86.30±14.59	75.22±14.23	0.857±0.085	87.43±12.15	83.22±14.52	0.421±0.034	87.43±10.06	83.22±13.30	0.575±0.021
class14	92.02±6.52	77.26±9.52	1.023±0.089	93.64±5.98	72.36±9.69	1.230±0.072	91.01±6.68	76.24±9.34	0.876±0.132	91.01±5.92	76.24±8.30	0.414±0.122
class15	83.41±4.72	83.66±6.25	0.620±0.085	86.79±3.69	80.56±8.96	0.416±0.052	88.00±4.71	82.46±8.38	0.364±0.062	88.00±1.20	82.46±2.51	0.752±0.032
class16	73.55±4.56	63.21±6.52	1.026±0.112	76.91±5.63	69.32±6.23	0.476±0.125	75.92±5.77	66.35±9.14	0.562±0.106	75.92±4.47	66.35±7.89	0.663±0.042
Mean	65.90±5.62	53.21±6.05	0.954±0.102	66.55±5.02	51.01±6.52	1.023±0.095	67.03±4.98	53.52±5.63	0.966±0.0895	**76.26±1.90**	**68.70±1.98**	**0.689±0.053**

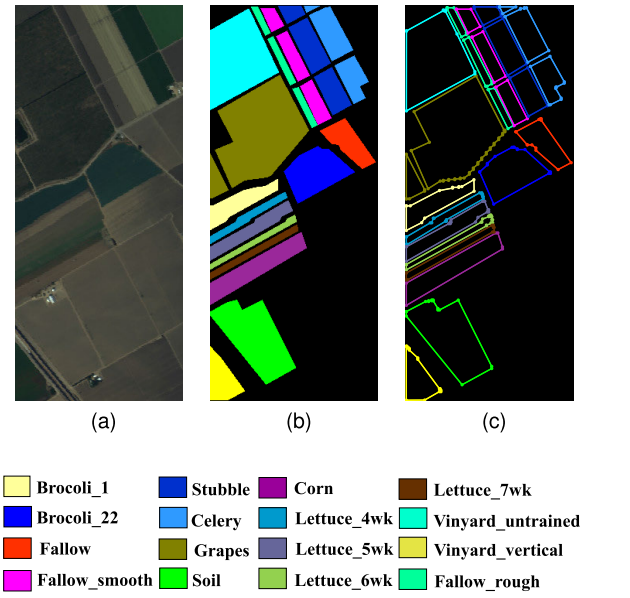


Fig. 6. Salinas valley dataset. (a) Three-band false color composite. (b) Original ground truth. (c) Vector ground truth.

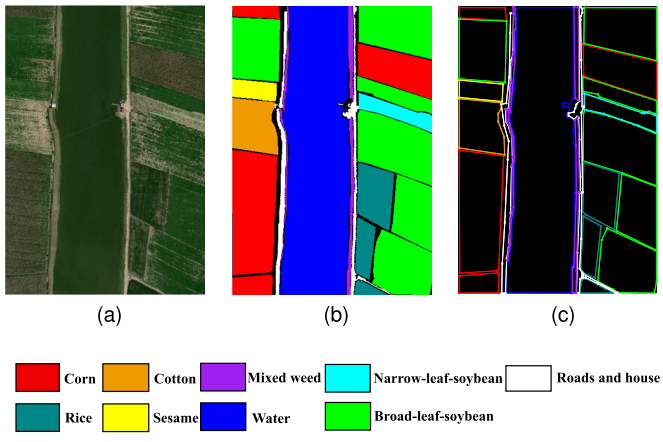


Fig. 7. WHU-Hi-LongKou dataset. (a) Three-band false color composite. (b) Original ground truth. (c) Vector ground truth.

ground-truth information comprises about 90% of the pixels (173 582 of 195 980) labeled into 19 different classes. For details of each category, see Fig. 8.

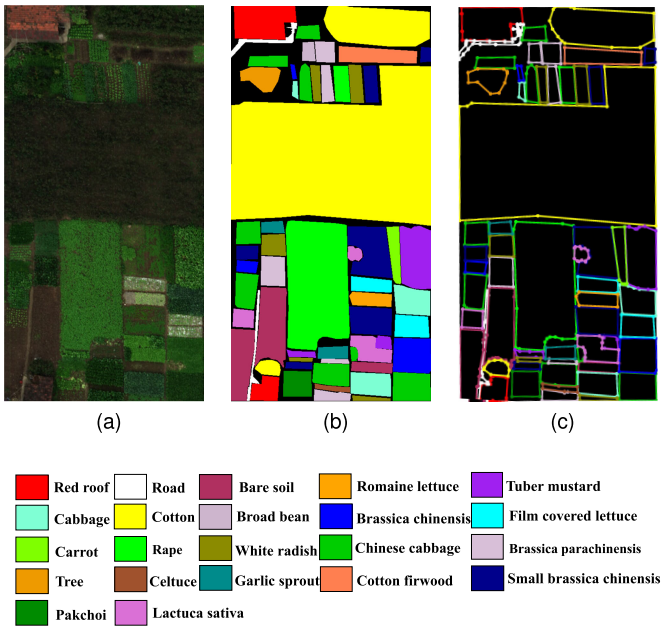


Fig. 8. WHU-Hi-HongHu dataset. (a) Three-band false color composite. (b) Original ground truth. (c) Vector ground truth.

B. Evaluation Metrics

To evaluate the performance of the proposed method, we use the typical index intersection over union (IOU) to reflect the accuracy of HSI vectorization compared with raster data. In addition, to directly compare the quality of the predicted vector, we report the evaluation of PoLiS metric [53]. For two given vectors, A and B, the PoLiS is defined as the average distance between each vertex. Vector A has n junctions $a_j \in A, j = 1, \ldots, n$ of A and its closest point b within the boundary of B that has m junctions $b_k \in B, k = 1, \ldots, m$. The PoLiS metric can be expressed as

$$\text{PoLiS}(A, B) = \frac{1}{2n} \sum_{a_j \in A} \min_{b \in \partial B} \|a_j - b\| + \frac{1}{2m} \sum_{b_k \in B} \min_{a \in \partial A} \|b_k - a\| \quad (8)$$

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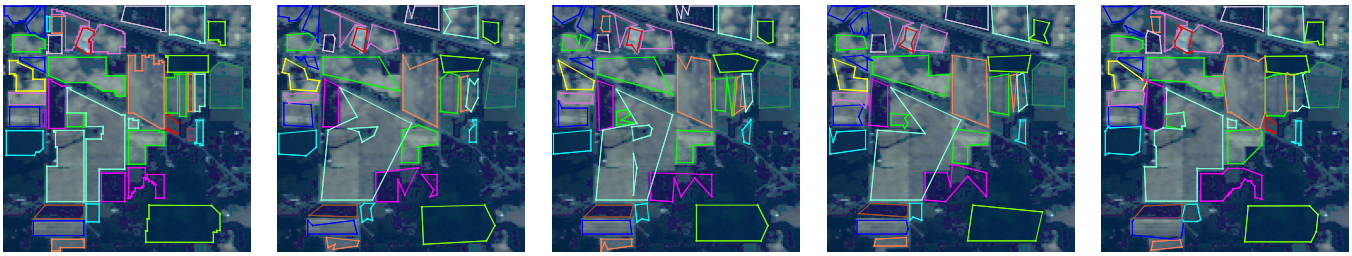


Fig. 9. Vectorization results for the Indian Pines dataset obtained by (a) Ground truth, (b) SpectralFormer [29] + DP, (c) MorphFormer [34] + DP, (d) HRNet [49] + DP, and (e) HSI-VecNet.

where $(1/2n)$ and $(1/2m)$ are the normalization factors. The IoU threshold for filtering the predicted vector is set to 0.5 [54].

The complexity aware IoU (C-IoU) [22] are also calculated for evaluation. For the two vectors A and B, the C-IoU metric is computed as

$$\text{C-IoU}(A, B) = \text{IoU}(A_m, B_m) \cdot (1 - \text{RD}(N_A, N_B)) \quad (9)$$

where the first term $\text{IoU}(\cdot)$ indicates the normal IoU between two compared vectors mask A_m and B_m. The second term $\text{RD}(N_A, N_B) = |N_A - N_B|/(N_A + N_B)$ is the relative difference between the total number of vertices N_A from vector A and the number of vertices N_B from vector B. The C-IoU metric considers the accuracy of the segmentation mask and the complexity of the vector, which reflects the matching degree of the number of vector junctions.

C. Parameter Setting

For the Indian Pines dataset and Salinas Valley dataset, we use SGD as the optimizer and set the base learning rate to 0.1. For the WHU-Hi-LongKou dataset and WHU-Hi-HongHu dataset, we use the Adam optimizer and set the learning rate to 0.0005. The total number of train cycles is set as 800 epochs, for every 100 epochs, and the learning rate is multiplied by 0.9. Moreover, the ratios of training samples are set as 20%, which is what we found in the ablation experiment. Our experiments are conducted on NVIDIA GeForce RTX 3090 and the deep learning platform is PyTorch (version 1.7.0).

D. Results and Analysis

In this section, to verify the performance of the HSI-VecNet, four vectorization HSI datasets are employed. Since the proposed method is the first end-to-end vectorization work in the HSI field, we have to choose some recent deep network-based HSI classification methods with post-process to compare, including SpectralFormer [29] and MorphFormer [34]. At present, DP [20] is the most common vectorization post-processing method, which has been used as the baseline in the field of natural image vectorization. Therefore, it is taken as the compared post-process in this section. Moreover, we also compare the same backbone network with post-process (HRNet + DP).

1) Performance on Indian Pines: Table I shows the quantitative comparison results of the Indian Pines dataset. Benefiting from the SSJP module, the performance of HSI-VecNet is higher than those of the post-processing-based methods. For instance, HSI-VecNet is 12.1% and 22.1% better than the same HRNet backbone in IoU and C-IoU. The results show that HSI-VecNet can obtain vector results with higher precision masks that have better raster-vector reversibility. Meanwhile, the PoLiS of HSI-VecNet is lower than other methods, which indicates the results obtained by HSI-VecNet are more concise and have a higher quality of the junction locations. Besides, it can be seen that the IoU and C-IoU of some categories are zero. This is because the area of these categories in the dataset is too small to be converted to effective vector results.

From the visualization perspective, Fig. 9 gives the vectorization results of different approaches on the Indian Pines. The visual effects produced by HSI-VecNet are better than DP-based methods. Especially, the advantages of HSI-VecNet in vector complexity are more obvious. DP-based methods are seriously affected by hyperparameters and more manual adjustments on vector quality. When the target size distribution in HSI is unbalanced, it is usually unable to accurately retain key junctions, resulting in a large difference between the simplified vector results and the ground truth. In our proposed HSI-VecNet, the junctions predicted by SSJP can be directly used to preserve the shape accuracy of the vector. In short, these comparisons demonstrate the excellent performance of HSI-VecNet on this dataset.

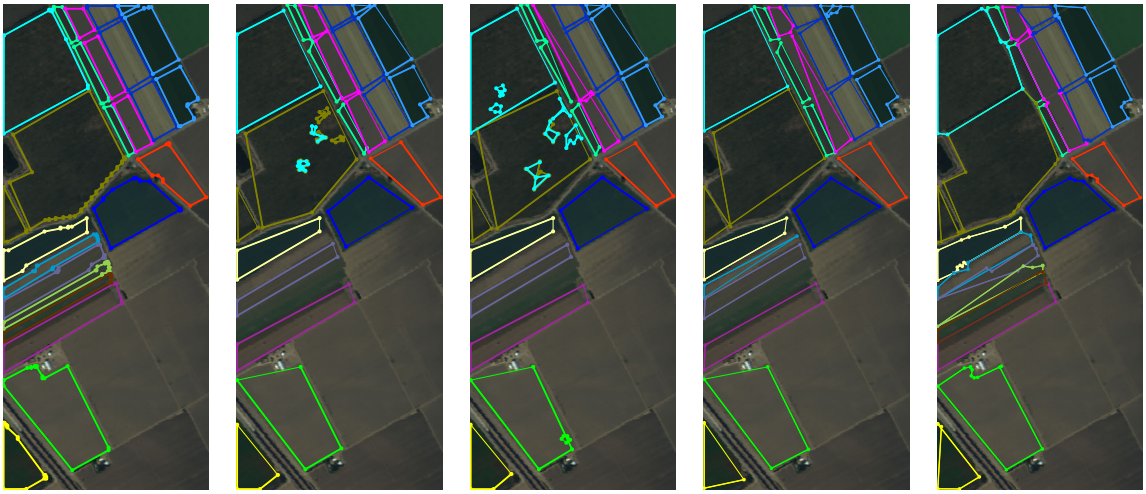
2) Performance on Salinas Valley: Table II compares the experimental results of HSI vectorization to various classes on the Salinas Valley dataset. From Table II, HSI-VecNet achieves the best accuracy (IoU of 80.30%, C-IoU of 61.69%, and PoLiS of 1.296) among the methods. The vectorization performance of SpectralFormer and MorphFormer are close on IoU, C-IoU, and PoLiS. Since post-processing-based methods often rely on manual parameter adjustment, extracting the best vectors from the classification mask is difficult. From the visualization perspective, Fig. 10 shows the vectorization results of different approaches in the Salinas Valley. As can be seen, the vectorization results of the SpectralFormer and MorphFormer exhibit some apparent errors in vectors due to the accumulation of classification errors in the pipeline. Meanwhile, HSI-VecNet can reduce the impact of the classification results' errors and maintain the vector's appropriate complexity.

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TABLE II

COMPARISON RESULTS OF DIFFERENT HSI VECTORIZATION METHODS ON THE SALINAS VALLEY DATASET, INCLUDING PER-CLASS IOU, C-IOU, AND POLIS

Category	SpectralFormer [29] + DP			MorphFormer [34] + DP			HRNet [49] + DP			HSI-VecNet		
	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓
class1	86.23±6.26	40.22±0.86	2.452±0.105	88.63±6.03	35.62±9.02	2.203±0.092	87.14±1.54	38.73±6.81	1.826±0.136	87.30±6.18	81.19±5.36	1.036±0.123
class2	86.02±4.56	37.62±7.36	2.032±0.121	86.52±3.95	40.36±7.62	1.632±0.162	87.05±3.05	38.55±8.94	1.856±0.121	95.59±2.90	66.62±3.07	2.056±0.101
class3	90.30±2.78	40.26±6.22	2.632±0.098	90.20±2.63	41.23±5.69	2.362±0.122	91.23±3.00	42.07±4.01	2.342±0.091	93.13±2.31	66.54±5.50	1.895±0.089
class4	22.26±16.29	16.00±9.36	1.451±0.075	33.02±17.36	13.62±9.65	1.875±0.193	23.29±14.20	14.92±7.35	1.625±0.175	70.94±3.62	63.47±6.65	0.985±0.078
class5	59.06±16.52	36.53±16.23	1.638±0.112	65.20±12.24	30.25±12.36	2.036±0.203	66.73±14.40	34.84±14.27	1.526±0.103	79.67±4.73	69.87±9.50	1.106±0.118
class6	76.32±5.36	68.92±16.53	1.295±0.084	80.36±5.62	68.52±18.62	1.524±0.075	85.32±5.21	70.18±16.31	0.526±0.079	82.12±5.16	75.45±7.05	0.657±0.196
class7	86.33±5.89	62.59±18.59	2.360±0.145	85.36±2.56	57.63±12.36	2.541±0.119	84.89±7.06	58.84±11.84	1.471±0.195	85.88±6.92	67.43±10.76	1.240±0.235
class8	78.92±4.52	22.36±5.56	3.652±0.162	80.36±4.23	20.36±5.69	3.062±0.156	82.15±5.26	21.95±6.00	2.562±0.133	93.72±1.09	57.22±5.78	1.725±0.178
class9	91.20±3.61	35.62±8.95	2.781±0.132	82.62±3.02	35.55±8.69	2.754±0.213	89.05±3.20	37.70±9.39	1.895±0.102	93.33±1.26	64.99±8.50	1.424±0.145
class10	85.53±20.65	60.25±19.46	2.654±0.142	88.51±21.41	62.57±2.02	1.645±0.139	87.18±3.09	63.40±2.25	1.562±0.143	74.67±24.38	69.02±26.36	0.527±0.163
class11	49.33±10.00	15.20±10.34	2.036±0.075	40.62±10.36	12.69±13.36	2.115±0.185	46.99±14.11	13.43±10.18	2.465±0.149	66.73±9.86	42.34±9.01	1.645±0.084
class12	78.29±20.36	30.25±13.95	2.874±0.068	80.41±3.89	35.66±12.55	2.326±0.177	79.38±13.89	31.75±13.56	1.877±0.095	67.74±19.04	46.97±12.26	0.936±0.095
class13	42.36±12.02	12.36±7.22	2.856±0.162	38.20±16.36	8.68±6.30	2.162±0.136	39.31±18.86	10.26±7.53	1.564±0.056	55.63±6.11	30.68±6.11	1.474±0.152
class14	25.67±8.56	15.20±11.85	1.623±0.097	31.63±29.62	15.96±12.36	1.895±0.145	29.83±18.70	13.77±13.25	2.506±0.162	60.45±7.57	47.58±7.07	1.129±0.124
class15	92.21±2.36	55.85±1.65	1.962±0.126	92.55±2.49	58.62±2.36	1.963±0.095	93.24±2.51	53.28±1.44	1.965±0.152	93.50±1.59	88.71±4.26	0.941±0.115
class16	85.61±3.69	33.20±8.58	3.062±0.185	81.50±4.52	30.26±8.55	2.023±0.185	88.74±5.72	31.12±5.76	2.231±0.132	85.83±1.63	47.76±7.13	1.865±0.130
Mean	70.98±5.02	36.4±7.65	2.346±0.156	71.61±6.03	35.47±8.67	2.120±0.178	72.59±5.98	35.92±6.95	1.862±0.135	**80.30±1.89**	**61.69±1.78**	**1.296±0.122**



(a) (b) (c) (d) (e)

Fig. 10. Vectorization results for the salinas valley dataset obtained by (a) Ground truth, (b) SpectralFormer [29] + DP, (c) MorphFormer [34] + DP, (d) HRNet [49] + DP, and (e) HSI-VecNet.

TABLE III

COMPARISON RESULTS OF DIFFERENT HSI VECTORIZATION METHODS ON THE WHU-HI-LONGKOU DATASET, INCLUDING PER-CLASS IOU, C-IOU, AND POLIS

Category	SpectralFormer [29] + DP			MorphFormer [34] + DP			HRNet [49] + DP			HSI-VecNet		
	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓
class1	92.94±1.34	72.28±1.67	1.214±0.151	93.62±1.31	80.23±2.35	0.896±0.154	95.28±1.60	80.23±1.85	0.903±0.144	97.99±0.48	95.42±1.66	0.962±0.073
class2	90.68±0.88	64.77±0.56	1.456±0.162	92.36±1.02	57.97±0.46	1.325±0.195	94.20±0.74	56.32±0.46	1.365±0.156	97.46±0.43	87.71±0.39	1.465±0.096
class3	92.64±2.65	82.35±3.04	0.658±0.052	93.02±3.58	82.93±2.82	0.859±0.095	93.30±3.17	84.20±2.82	0.845±0.078	92.98±1.34	84.53±1.22	0.151±0.102
class4	91.46±0.86	53.98±4.12	1.562±0.179	90.36±0.34	57.69±18.38	1.795±0.201	92.32±0.57	57.03±28.38	1.654±0.125	96.85±0.15	67.00±26.32	1.789±0.080
class5	87.80±1.33	39.02±7.63	1.785±0.095	88.63±3.78	38.35±0.65	1.986±0.185	86.29±4.46	35.36±0.65	1.562±0.117	93.51±3.29	88.90±4.96	0.923±0.045
class6	93.09±1.62	69.82±4.02	1.657±0.084	95.62±1.63	68.64±3.44	1.685±0.122	94.56±1.05	71.62±3.44	1.789±0.189	91.86±0.68	69.68±2.95	1.267±0.062
class7	92.08±0.87	11.51±5.12	2.689±0.165	91.67±0.48	11.62±0.05	2.748±0.131	92.95±0.90	21.51±3.69	2.032±0.168	98.11±0.43	95.34±3.24	0.116±0.113
class8	0±0	0±0	—	0±0	0±0	—	0±0	0±0	—	56.99±1.83	49.67±1.16	1.245±0.085
class9	0±0	0±0	—	0±0	0±0	—	0±0	0±0	—	54.70±1.78	43.75±1.63	1.442±0.103
Mean	71.19±3.75	43.75±4.62	1.574±0.156	71.70±3.62	44.15±5.03	1.613±0.178	72.10±4.98	45.14±6.95	1.450±0.135	**86.72±0.24**	**75.78±2.91**	**1.043±0.083**

3) Performance on WHU-Hi-LongKou: Table III summarizes the comparative results on the WHU-Hi-LongKou dataset, and Fig. 11 shows the corresponding vectorization results. We can see from Fig. 7 that the target distribution is more dense than Indian Pines and Salinas Valley datasets. Significantly, some targets are even indistinguishable from each other. Accordingly, some categories in this dataset are challenging to extract regular vector shapes, resulting in the poor performance of IoU and C-IoU, such as roads and houses. Table III shows that the results of the DP-based methods on IoU are slightly lower than those generated by HSI-VecNet, but IoU can only partly evaluate the quality of the vectors. In the C-IoU metric, the DP-based methods perform poorly in each category, which means that many junctions are not retained or are redundant. The vector results have a big gap with the actual situation. Experimental results show that learning junction locations is the key to successfully achieving the HSI vectorization task. Unlike post-

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processing methods, HSI-VecNet uses a two-branch structure to learn semantic information and junction locations simultaneously. This structure can obtain vector junctions by deep learning rather than simplifying outlines by artificial distance constraints. In contrast, HSI-VecNet performs better in C-IoU and PoLiS, which indicates that the method proposed in this article can better complete the HSI vectorization task.

4) Performance on WHU-Hi-HongHu: The quantitative results are listed in Table IV, and the corresponding visual vectorization results on the WHU-Hi-HongHu dataset are shown in Fig. 12. Compared with the previous three datasets, this dataset has a higher spatial resolution and denser target distribution. In Table IV, it can be found that the DP-based methods have higher PoLiS, which suggests that the post-processing methods cannot be adapted to complex geometric shapes. Benefiting from the junction prediction branch, HSI-VecNet can obtain shape-priority knowledge from the classification task and adapt to more complicated goals. In contrast, HSI-VecNet is about 10% higher than other methods on IoU, proving that this method can maintain high reversibility between vector and mask. At the same time, the C-IoU of HSI-VecNet is about 20% higher than others, which indicates that HSI-VecNet can make full use of the information between the two branches. The vector shape of HSI-VecNet is more accurate while maintaining high reversibility. The above analysis shows that HSI-VecNet is effective in this dataset and has better vector extraction performance.

5) Performance Analysis: Our method is compared with other post-processing-based methods on the four popular datasets, and we find that HSI-VecNet obtained better performance in all metrics. Especially our approach has the most significant improvement in C-IoU, which proves HSI-VecNet can extract more accurate junctions with instance-level semantic information guidance. Mainly, the post-processing-based methods can only extract vector junctions from the segmentation masks by fixed constraints. Regarding the visual comparison in Figs. 9–12, we can see that the post-processing based approaches are easy to ignore critical junctions, resulting in the wrong vector shapes. As can be seen that our HSI-VecNet performs better in the WHU-Hi-LongKou dataset that has more complex geometric shapes. In the WHU-Hi-LongKou dataset, several objects have extremely irregular shapes and are indistinguishable from each other. The proposed HSI-VecNet explores the instance and geometry information to adaptively extract junctions, thus having lower PoLiS on all datasets. For Indian Pines, Salinas Valley, and WHU-Hi-HongHu datasets, as can be noticed that the improvement of HSI-VecNet in PoLiS and C-IoU is significantly higher than that in IoU. This is because the above three datasets are less sensitive to the parameters in post-processing methods due to the low complexity of shapes. So, the advanced performance of the HSI-VecNet is more reflected in the precise matching of junctions. The above analyses indicate that HSI-VecNet can achieve a significant improvement in HSI vector extraction by leveraging the spectral–spatial information.

E. Ablation Studies

1) Ablation Study of CSSAM: In HSI-VecNet, we use CSSAM to exchange information on instance extraction and junction prediction branches. We integrate semantic and geometric features to achieve more accurate junction prediction and extraction of different instances and obtain higher quality HSI vectorization results.

To verify the effectiveness of the module, an ablation experiment is designed to ascertain the effect of CSSAM. To this end, we use the baseline to represent the model structure with only two branches without information interaction and compare the complete HSI-VecNet using the CSSAM and the post-processing method. Table V shows the relevant experimental results on the Indian Pines dataset.

Experimental results showed that baseline is slightly better than the DP method on IoU and PoLiS but slightly lower on C-IoU, possibly due to the loss of semantic-level feature information leading to inaccurate junction prediction. What's more, the use of CSSAM can significantly improve the model performance. More specifically, adding the CSSAM improves the IoU, C-IoU, and PoLiS scores on the WHU-Hi-HongHu by 13.95%, 32.43%, and 1.816%.

2) Ablation Study of the Percentage of Train Samples: In this section, we test the classification accuracy of HSI-VecNet proposed in this article under different numbers of labeled samples. Different from the general HSI classification task, we also need to distinguish whether each pixel belongs to the unmarked category during instance extraction. So, the number of classes is one more than the known categories during classification. Due to the increasing classification difficulty, we set the proportion of training samples at 5%–25% for each category and took IoU as the measurement index. It can be seen that the performance of the proposed HSI-VecNet on the four datasets increases monotonically with the increase of training samples, as shown in Fig. 13. With the increase in training samples, HSI-VecNet will gradually reach saturation in the four datasets. With the continuous increase of training samples, the performance will improve little. In the Indian Pines and Salinas Valley datasets, HSI-VecNet reaches saturation at about 20%, while in WHU-Hi-LongKou and WHU-Hi-HongHu, are about 20%–22%, respectively. To sum up, the training samples required by HIS-VecNet should be controlled at about 20% of each category. In this article, the labeling of 20% is adopted for all four datasets, which can maintain high accuracy and save some labeling costs, which is more in line with the actual needs.

3) Ablation Study of the Loss Factors: In the method part, our loss function consists of three parts, which are L_{cla}, L_{vcla}, and L_{voff}. To balance these different loss functions, we use three loss factors to solve it, including λ_{cla}, λ_{vcla}, and λ_{voff}. Setting these loss factors is mainly depending on two principles. The first design principle is to balance the loss value on the same order of magnitude. We took the convergent values of the three loss functions as the reference values when they were separately trained. The second design principle is that the loss factor value's size determines the loss function's importance. Since the size of L_{voff} is not sensitive to our

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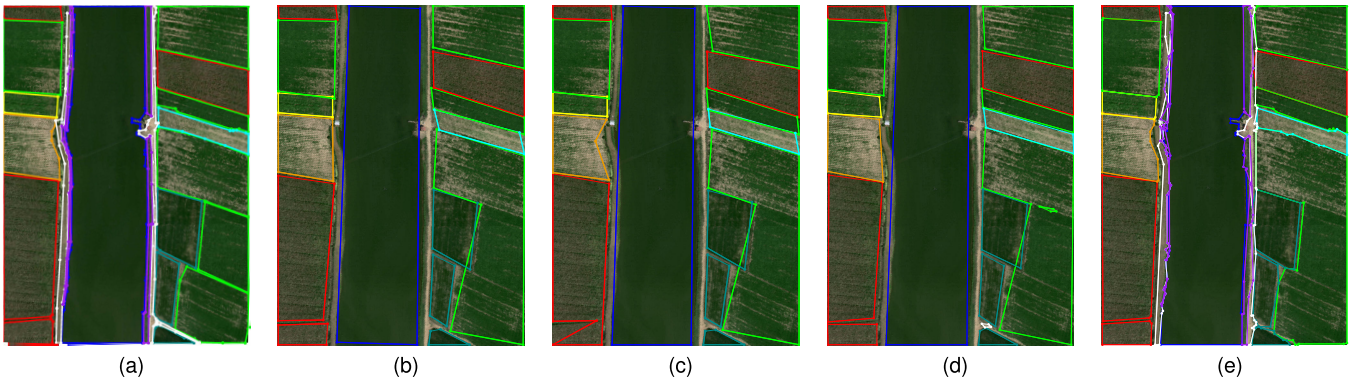


Fig. 11. Vectorization results for the WHU-Hi-LongKou dataset obtained by (a) Ground truth, (b) SpectralFormer [29] + DP, (c) MorphFormer [34] + DP, (d) HRNet [49] + DP, and (e) HSI-VecNet.

TABLE IV

COMPARISON RESULTS OF DIFFERENT HSI VECTORIZATION METHODS ON THE WHU-HI-HONGHU DATASET, INCLUDING PER-CLASS IOU, C-IOU, AND POLIS

Category	SpectralFormer [29] + DP			MorphFormer [34] + DP			HRNet [49] + DP			HSI-VecNet		
	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓	IoU↑	C-IoU↑	PoLiS↓
class1	96.45±1.63	45.39±5.78	1.802±0.113	91.23±1.32	47.56±1.03	2.031±0.126	93.25±1.77	43.88±8.83	2.132±0.063	95.91±0.62	90.05±4.54	1.012±0.012
class2	37.77±3.54	4.62±1.22	3.652±0.085	44.52±3.25	27.63±2.65	1.892±0.095	48.12±2.10	8.91±4.77	2.956±0.152	66.46±3.07	55.76±1.08	0.865±0.036
class3	88.80±0.44	36.74±6.25	2.347±0.132	86.32±3.41	40.20±6.32	2.356±0.086	89.87±7.40	39.60±6.18	1.685±0.175	91.22±1.73	81.71±5.25	1.321±0.045
class4	83.14±1.12	41.57±3.62	2.152±0.115	91.63±1.62	52.36±4.59	2.136±0.133	94.85±1.83	61.28±3.53	1.965±0.095	96.34±0.93	86.17±3.35	0.657±0.015
class5	0±0	0±0	—	57.21±17.52	43.00±18.62	1.263±0.142	67.09±21.64	60.31±22.27	1.142±0.084	88.67±2.10	73.62±12.15	0.741±0.057
class6	90.46±1.78	48.25±3.97	1.854±0.854	93.26±1.23	45.28±3.26	2.156±0.102	92.32±0.69	50.12±2.27	2.132±0.251	95.98±0.32	72.39±2.71	1.342±0.036
class7	85.53±2.86	63.13±5.56	1.123±0.751	88.23±2.64	60.85±5.65	1.759±0.084	86.78±3.53	68.55±3.05	1.765±0.305	90.23±1.61	74.89±4.59	0.859±0.087
class8	95.38±3.62	84.79±2.22	0.895±0.621	83.26±2.54	75.62±2.15	1.354±0.043	86.88±2.64	77.22±9.35	1.652±0.041	85.65±21.13	72.16±15.15	0.414±0.065
class9	93.81±5.54	72.63±6.62	1.546±0.152	88.20±3.32	65.84±7.63	2.046±0.051	91.91±11.49	71.87±2.08	2.036±0.081	91.96±2.32	71.82±5.47	1.112±0.014
class10	88.20±3.85	47.04±2.56	1.856±0.135	91.30±0.66	47.96±1.44	2.564±0.162	89.95±5.76	48.55±6.49	2.121±0.175	96.49±0.37	93.35±1.73	0.225±0.036
class11	81.22±1.63	59.07±4.96	1.862±0.265	83.26±2.14	70.42±5.32	1.541±0.362	85.76±2.49	73.47±3.68	1.785±0.125	90.89±0.80	78.59±1.86	0.782±0.057
class12	88.57±2.56	85.72±3.62	1.036±0.254	92.05±3.01	75.63±6.05	1.895±0.241	90.76±2.76	80.68±3.72	1.695±0.021	93.69±1.15	81.39±5.86	0.741±0.049
class13	89.46±2.55	75.03±12.56	1.564±0.079	91.63±0.98	83.25±8.65	1.598±0.178	90.36±3.71	79.06±11.31	2.362±0.125	93.49±0.90	71.77±9.69	1.235±0.026
class14	87.17±1.78	51.27±4.62	1.562±0.074	87.25±0.65	53.66±0.64	2.147±0.195	89.01±6.89	52.36±10.52	2.651±0.112	91.20±0.86	85.40±3.54	0.658±0.042
class15	76.14±18.32	53.37±10.23	2.354±0.165	25.32±4.26	6.32±16.62	2.046±0.124	6.29±14.06	3.77±8.43	2.340±0.136	14.11±25.75	11.73±24.45	1.225±0.051
class16	93.72±1.44	78.92±4.12	1.805±0.095	72.13±2.89	50.32±8.25	1.963±0.135	88.57±11.48	74.59±16.25	1.745±0.084	94.60±0.38	85.40±3.45	0.785±0.096
class17	78.73±8.32	62.98±14.88	2.324±0.084	80.26±6.52	73.62±15.26	1.248±0.212	82.74±7.93	76.93±13.97	1.254±0.095	96.00±1.87	93.10±5.61	1.032±0.085
class18	63.95±16.88	31.98±14.21	3.512±0.154	55.63±15.20	28.62±11.36	2.341±0.215	61.88±17.58	34.60±13.56	2.142±0.062	94.59±2.31	88.94±5.90	0.906±0.064
class19	85.05±2.56	71.21±3.52	1.965±0.123	82.74±3.65	75.62±6.52	1.135±0.062	81.38±2.87	70.29±8.48	1.639±0.162	90.55±1.84	76.20±5.38	1.421±0.067
class20	86.14±1.35	76.57±6.30	1.562±0.074	80.36±2.85	77.53±5.30	1.368±0.854	81.64±1.27	78.02±5.30	1.952±0.152	91.90±2.19	80.84±4.78	1.652±0.038
class21	89.05±22.95	79.16±20.30	1.874±0.122	60.85±10.45	58.65±8.62	1.027±0.142	61.86±23.86	56.94±22.20	1.286±0.113	77.85±36.56	74.26±40.14	0.512±0.051
class22	58.05±6.81	30.96±14.06	2.302±0.062	80.62±6.55	60.25±5.30	2.120±0.174	81.65±7.33	62.58±14.01	2.234±0.095	96.38±0.80	88.43±9.16	0.324±0.048
Mean	78.80±4.02	54.56±3.96	1.950±0.112	77.60±4.52	55.46±5.21	1.820±0.151	79.22±4.64	57.89±4.95	1.940±0.135	**87.46±1.02**	**76.73±1.73**	**0.904±0.045**

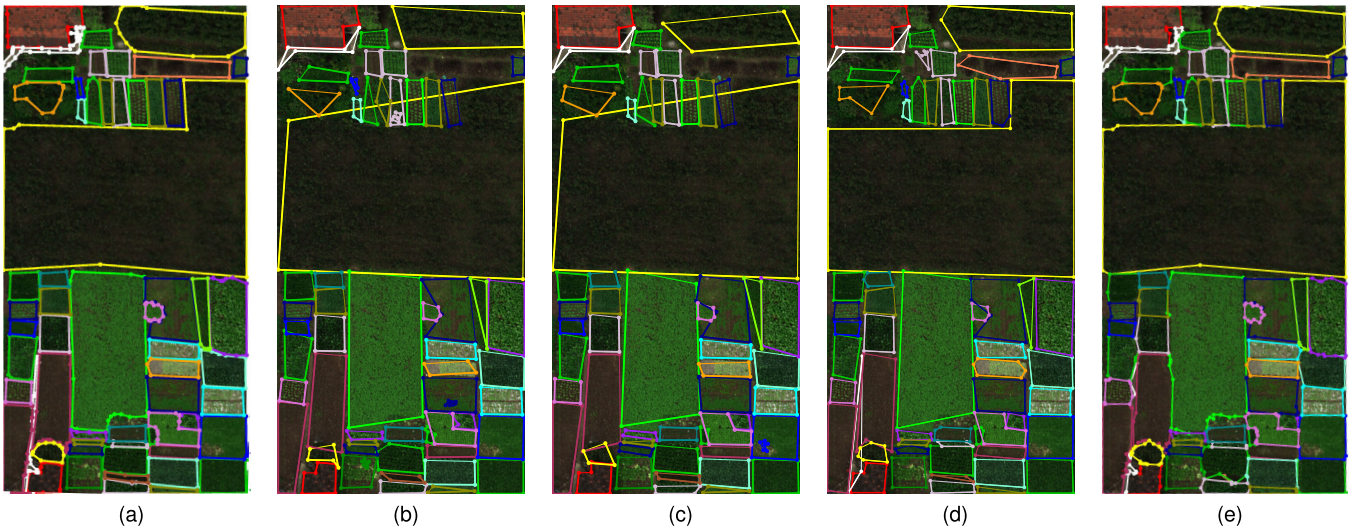


Fig. 12. Vectorization results for the WHU-Hi-HongHu dataset obtained by (a) Ground truth, (b) SpectralFormer [29] + DP, (c) MorphFormer [34] + DP, (d) HRNet [49] + DP, and (e) HSI-VecNet.

method, which only was used to fix the junction locations to realize the sub-pixel level. Based on the magnitude of L_{voff}, we set the value of λ_{voff} to a fixed value of 0.25. So we mainly explore the effects of the ratio of L_{cla} and L_{vcla} on the model. From the experimental results in Table VI, it can be seen that the performance of the model reaches the best when λ_{cla}:λ_{vcla}:λ_{voff} = 4:8:0.25. Therefore, we choose λ_{cla}:λ_{vcla}:λ_{voff} = 4:8:0.25 as our loss factor ratios.

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TABLE V
ABLATION STUDY FOR THE CSSAM ON THE WHU-HI-HONGHU

Metric	IoU↑	C-IoU↑	PoLiS↓
HRNet + DP	70.06	51.31	3.04
Baseline	73.51	44.30	2.72
Baseline + CSSAM	**87.46**	**76.73**	**0.904**

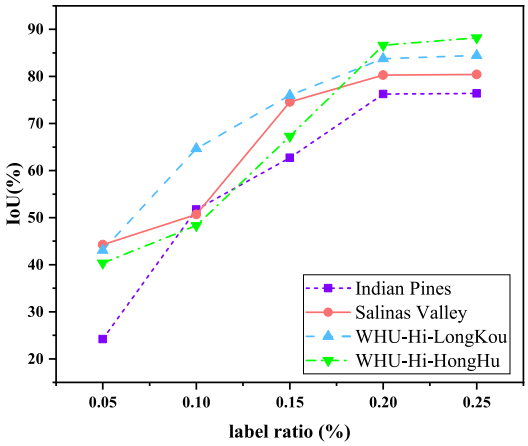


Fig. 13. Vectorization performance of HSI-VecNet with different training set ratios on four datasets.

TABLE VI
ABLATION STUDY FOR THE LOSS FACTORS ON THE WHU-HI-HONGHU

$\lambda_{cla} : \lambda_{vcla} : \lambda_{voff}$	IoU↑	C-IoU↑	PoLiS↓
1 : 8 : 0.25	73.88	47.15	1.588
2 : 8 : 0.25	79.16	61.72	1.762
4 : 8 : 0.25	**87.46**	**76.73**	0.904
8 : 8 : 0.25	85.36	76.70	**0.897**
8 : 4 : 0.25	86.52	76.50	0.956
8 : 2 : 0.25	86.08	76.14	0.997
8 : 1 : 0.25	86.54	72.63	1.027

V. CONCLUSION

This article introduced vectorization into HSI image interpretation for the first time. To support this, we labeled four popular HSI datasets in vector form and constructed four HSI vectorization benchmark datasets. Based on this foundation, we proposed a new method for end-to-end HSI vectorization called HSI-VecNet. The proposed HSI-VecNet uses two branches of instance extraction and junction prediction for information exchange, improving traditional post-processing methods and producing vector results of higher quality. Experimental results demonstrate that HSI-VecNet outperforms the state-of-the-art methods on all four datasets. In addition, to assess the effectiveness of the CSSAM in HSI-VecNet, we conducted ablation experiments. Our results reveal that the cross-task information interaction facilitated by the CSSAM has a promoting effect on the model's performance. HSI-VecNet can directly extract the vector results of different land types in HSI, which meets the vector data requirements in the practical application of geographic information system (GIS). In the future, we will delve deeper into the field of HSI vectorization, aiming to create a more effective HSI-VecNet that can better accommodate the demands of real-world scenarios characterized by limited data and intricate categorical variations.

In the future, we will further explore the HSI vectorization method with fewer train data and higher reasoning speed, which will be important for the satellite and airborne platforms. We hope the proposed method will serve as a strong baseline and aid future research in HSI vectorization. In addition, we will investigate how to simultaneously detect vector structures and perform panoramic segmentation to achieve a better understanding of hyperspectral remote sensing.

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